MILITARY SPECIFICATION

RECEIVER-TRANSMITTER RT-1184( )/GRA-114

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies by the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Receiver-Transmitter RT-1184( )/GRA-114 which is used as the microphone position of Radio Data Link, Sound Ranging AN/GRA-114 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Materials-handling (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock Container Grade and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1290

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MIL-P-116	Preservation-packaging, Methods of
MIL-S-3950	Switches, Toggle, General Specification for
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging

STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-678035	Burn-in Cycle
SM-A-865230	Performance Test Procedures

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 (see 6.2(g)).

3.2 Construction. The Receiver-Transmitter RT-1184( )/GRA-114 (Microphone Position) shall be constructed in accordance with the requirements of this specification and the model.

3.2.1 Knobs. Knobs on code switches shall be provided with key-ways that insure proper orientation on the switch shafts.

3.2.2 Gas emission. Electrical arcing shall be prevented in the electronics compartment of RT-1184. Battery compartments shall be isolated from the electronic compartments by means of air-tight and chemical-tight bulkheads.

3.2.3 Panel. The exterior connector panel shall be constructed to provide high visibility of the connectors to operating personnel and convenient connections to the panel, of wires and cables, by personnel wearing arctic gloves.

3.2.4 Printed circuit boards. Where feasible, printed circuit boards shall be provided with pull-out flanges. Flanges shall be labeled.

3.2.5 Test points. Test points shall be provided on all printed circuit boards. Test points shall be properly labeled and located so as to facilitate testing without removal of the boards from the chassis wherever feasible.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Toggle switches. Toggle switches, where employed, shall be full size in accordance with MIL-S-3950.

3.4 Finish. All surfaces of the RT-1184 required a protective coating shall be finished in accordance with MIL-F-14072 (see 4.4).

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3.5 Marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4). Front panel markings shall be group I as described in that specification. Shield cans shall be clearly marked to indicate the character and relative position of their interior contents.

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each RT-1184 shall be serial-numbered (see 4.4).

3.6 Interchangeability. Like units, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.6).

3.7 Burn-in. The RT-1184 shall meet the requirements of this specification after being subjected to the burn-in cycle of Drawing SC-A-678033 (see 4.7).

3.8 Performance requirements.

3.8.1 Transmitter.

3.8.1.1 Power output. The power output as measured under load at the external fitting for the antenna coaxial cable shall be in accordance with Table 1 (see 4.9).

TABLE I. Power output.

Temp. in°F	Temp. in°C	Battery(volts)	Power out(w) Minimum	Power Setting
-40° to +150° 1/	-40° to 65.6°	12-15 1/	3.15	High
+50° to +150°	10° to 65.6°	10	2.25	High
+50° to -40°	10° to -40°	10	2.25	High
-40° to +150°	-40° to 65.6°	12	0.25-1.0	Med

1/ Equipment to be capable of continuous operation under these conditions.

3.8.1.2 Efficiency. The efficiency shall be at least 27 percent (see 4.9).

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3.8.1.3 Duplexing loss. Duplexing filters shall not result in a power loss in excess of 2 dB as measured between the R.F. output terminals of the transmitter and the external antenna connector of RT-1184 (see 4.9).

3.8.1.4 Frequency. The frequency shall be within ±.007 percent of the assigned channel frequency (see 4.9).

3.8.1.5 Voice deviation. The FM deviation shall not be less than +7 KHz nor more than + 10 KHz when a 1000 Hz signal of 1 mv is applied to the input of the handset connector (see 4.9).

3.8.1.6 Deviation limiting. Deviation limiting shall not exceed +19.0 KHz when a 1.0 KHz signal of 10 mv is applied to the input of the handset connector (see 4.9).

3.8.1.7 Transmitter voice frequency response. The frequency response shall meet the requirements of Table II (see 4.9).

TABLE II. Voice frequency response.

Frequency (Hz)	Response (dB)
150	-3.0 +2.0
300	-1.0 ±2.0
600	0 ±1.0
1000	0 Reference
3500	-3.0 +2.0
5000	-6.0 min

3.8.1.8 Audio distortion. The audio distortion shall be less than 5 per cent with a carrier deviation of ± 8 KHz at a frequency of 1000 Hz (see 4.9).

3.8.1.9 Transmitter, data.

3.8.1.9.1 Dynamic range. The modulator shall accomodate incoming data signals in the range of 0.2- 20vP/P at the zero dB setting on the attenuator board. (S+N)/N shall be at least 10dB (see 4.9)

3.8.1.9.2 Data distortion The data distortion shall not exceed 4.5 percent with a carrier deviation of ± 10 KHz over a frequency range of 5 to 100 Hz (see 4.9).

3.8.1.9.3 Data frequency response. The data response shall be in accordance with Table III (see 4.9).

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TABLE III. Data frequency response.

Frequency (Hz)	Response (dB)
5	-3.0 +1, -2
10	-1 + .5
25	0 Reference
75	0 +1.0
100	0 ±1.0
150	-1.5 dB max
300	-5.0 dB max
600	-3.0 dB min
1000	-5.0 dB min

3.8.1.9.4 Data start up delay. The unit shall be fully stabilized and operational within 500 ms after the first keying signal arrives at the receiver (see 4.9).

3.8.1.10 Telemetry-operation. When either the internal batteries of RT-1184 drop to a level under load of 10.5V +0.5V, or the batteries of the M-161 Microphone drop to a level of 5.0V ±.25V the transmitter, when keyed, shall transmit a continuous 2.0 kHz ±25 per cent audio tone in addition to its normal modulating signals (see 4.9).

3.8.1.11 High voltage power supply. The high voltage power supply shall provide the voltages shown below, over the operating temperature range. Hum and noise shall not exceed 0.1 percent of the measured output voltage (see 4.9).

TABLE IV. Power Supply.

Battery supply (volts)	Output (volts)
10	90 minimum
12	112 ± 7
15	145 maximum

3.8.1.11.1 Protective circuit. The high voltage power supply shall be provided with overload protective circuitry. Overloads of short duration, as for example, current surges caused by the operator stomping on the M-161 microphone windshield shall not activate the protective circuitry (see 4.9).

3.8.1.11.2 Ripple. The ripple voltage under load shall not exceed 0.1 percent of the measured DC voltage at the output of the high voltage power supply (see 4.9).

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3.8.1.12 Fail-safe timer. If data transmission is not halted by a STOP command from the Command-Post, the RT-1184 shall be returned to standby condition by means of its built-in timer circuit within 4.5 to 6.0 minutes after receipt of its last Data START command (see 4.9). At cold temperatures return to standby shall not exceed 7 minutes.

3.8.2 Receiver.

3.8.2.1 Code sensitivity. The RF level required for triggering action shall be less than 1.0 uv. Only one train of code groups shall be necessary for triggering action (see 4.9).

3.8.2.2 Code capture ratio. The code capture ratio shall not exceed 3 dB, when measured at 5 mv and 100 mv signal strength (see 4.9).

3.8.2.3 Sensitivity. The sensitivity shall be such that 0.63 uv applied to the input terminals of the receiver shall produce a signal-plus-noise to noise ratio of at least 10 dB (see 4.9).

3.8.2.4 Limiting. Not more than 1.0 uv of input signal shall be required to achieve full limiting. The limiting characteristics shall be such that the audio power output shall not vary more than ± 3 dB at RF input levels of 1.0 to 100,000 uv (see 4.9).

3.8.2.5 Distortion. Receiver distortion at 5 mw output shall not exceed 5 percent (see 4.9).

3.8.2.6 Voice frequency response. The frequency response shall comply with Tabe V (see 4.9).

TABLE V. Voice frequency response.

Frequency (Hz)	Response (dB)
150	-3.0 +1.5
300	-1.0 ±1.0
600	0 ±1.0
1000	0 Reference
3500	-3.0 +2.0
5000	-5.0 min

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3.8.2.7 Audio output-power. The output level across 1,000 ohms at 12.0 vdc shall be at least 2.24 v rms and the output level at 10 vdc shall be at least 1.73 v rms (see 4.9).

3.8.2.8 Standby input-power. The standby current shall not be more than 25 ma (see 4.9).

3.8.2.9 Front-end overload. An interfering signal, 120 dB above 1 microvolt and removed +15 MHz from the desired frequency, shall not cause failure of any part of RT-1184 (see 4.9).

3.8.2.10 Duplexing desensitization. The receiver desensitization due to concurrent operation of the local transmitter in RT-1184 shall not exceed 2 dB as compared to the corresponding response to input signals exhibited by the receiver when the RT-1184 transmitter is in standby condition (see 4.9).

3.8.3 Instrumentation. The panel meter shall provide readings for the following parameters:

a. M-161 microphone current
b. Transmitter operation
c. M-161 battery condition under load
d. Internal battery condition under load

3.8.3.1 Meter readings. When the internal battery supply of RT-1184 is at a 10.0 V level and the M-161 Microphone battery is at a 5.0 V level the meter indicator shall fall into the following color bands:

Switch position	Meter reading
INT BAT	Green/Red Interface
MIC BAT	Green/Red Interface
MIC	Green
XMTR	Green

(see 4.9).

3.9 Service conditions. The RT-1184 shall meet the service conditions below.

3.9.1 Temperature.

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3.9.1.1 High temperature. During testing as specified in 4.10.1.1, the RT-1184 shall meet the following requirements:

Paragraph	Measurement
3.8.1.1	Power output
3.8.1.4	Frequency
3.8.1.5	Voice deviation
3.8.1.6	Deviation limiting
3.8.1.8	Audio distortion
3.8.1.9.1	Dynamic range
3.8.1.10	Telemetry operation
3.8.1.11	High voltage power supply
3.8.1.12	Fail-safe timer
3.8.2.1	Code sensitivity
3.8.2.3	Sensitivity
3.8.2.4	Limiting
3.8.2.5	Distortion
3.8.2.7	Audio output power

3.9.1.2 Low temperature. During testing as specified in 4.10.1.2, the RT-1184 shall meet the following requirements:

Paragraph	Measurement
3.8.1.1	Power output
3.8.1.4	Frequency
3.8.1.5	Voice deviation
3.8.1.6	Deviation limiting
3.8.1.8	Audio distortion
3.8.1.9.1	Dynamic range
3.8.1.10	Telemetry operation
3.8.1.11	High voltage power supply
3.8.1.12	Fail-safe timer
3.8.2.1	Code sensitivity
3.8.2.3	Sensitivity
3.8.2.4	Limiting
3.8.2.5	Distortion
3.8.2.7	Audio output power

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3.9.2 Humidity. The RT-1184 shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the following requirements during the operating times specified in 4.10.2:

Paragraph	Measurements
3.8.1.1	Power output
3.8.1.4	Frequency
3.8.1.5	Voice deviation
3.8.1.6	Deviation limiting
3.8.1.8	Audio distortion
3.8.1.9.1	Dynamic range
3.8.1.10	Telemetry operation
3.8.1.11	High voltage power supply
3.8.1.12	Fail-safe timer
3.8.2.1	Code sensitivity
3.8.2.3	Sensitivity
3.8.2.4	Limiting
3.8.2.5	Distortion
3.8.2.7	Audio output power

3.9.3 Immersion. The RT-1184 shall show no evidence of leakage after it is tested as specified in 4.10.3.

3.9.4 Dust. After testing as specified in 4.10.4, the RT-1184 shall meet full specification performance for the following measurements:

Paragraph	Measurements
3.8.1.1	Power output
3.8.2.1	Code sensitivity
3.8.2.7	Audio output power

In addition, all panel controls shall have unrestricted freedom of motion and all controls and fittings shall be capable of performing their intended function.

3.9.5 Vibration, bounce, and shock.

3.9.5.1 Bounce and shock. The RT-1184 shall meet the requirements of Table VI after submission to bounce and shock.

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TABLE VI Bounce and shock

Requirement Paragraph	Test Paragraph	Performance After Test
Bounce	4.10.5.1	Specified performance. 1/ No physical damage except minor surface abrasion. Transmitter shall not dislodge from mounting.
Shock; drop	4.10.5.2	Specified performance. 1/ Any physical damage shall be minor only. Transmitter shall not dislodge from mounting.

1/ The RT-1184 shall meet specified performance for the following:

Paragraph	Measurement
3.8.1.1	Power output
3.8.1.4	Frequency
3.8.1.5	Voice deviation
3.8.1.6	Deviation limiting
3.8.1.8	Audio distortion
3.8.1.9.1	Dynamic range
3.8.1.10	Telemetry operation
3.8.1.11	High voltage power supply
3.8.1.12	Fail-safe timer
3.8.2.1	Code sensitivity
3.8.2.3	Sensitivity
3.8.2.4	Limiting
3.8.2.5	Distortion
3.8.2.7	Audio output power

3.9.6 Rain. After testing as specified in 4.10.6, there shall be no evidence of water penetration into the electronics compartment.

3.9.7 Salt fog. After the salt fog test of 4.10.7, the RT-1184, when examined visually, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface. Electrical operation is not required.

3.9.8 Fungus. After the fungus test of 4.10.8, the RT-1184 shall be visually examined and shall reveal no evidence of viable fungus on the surface.

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3.10 Electromagnetic interference characteristics. The RT-1184 shall meet the emission and susceptibility requirements for MIL-STD-461 for Class I equipments. Test set-ups and test procedures shall conform to the measurement techniques of MIL-STD-462 (see 4.11). The following requirements are applicable:

Transmit Mode	Receive Mode	
CE06 3/	CS03	CE06
RE02	CS04 2/	RE02 1/
RS03 1/	CS05	RS03 1/

1/ The upper frequency of the test shall be 1000 Mhz.

2/ All responses shall be 96dB down except for a maximum of five (5) which shall not be less than 80dB down, when tested in accordance with MIL-STD-462.

3/ a. The equipment shall meet the spurious and harmonic output requirements of MIL-STD-461A, Notice 4, except for a total of three responses which may be between 70 and 80dB down. No response will be less than 70dB down relative to the fundamental power in the medium power mode.

b. The equipment shall meet the spurious and harmonic output requirements of MIL-STD-461A, Notice 4, except for a total of three responses which may be between 70 and 80dB down. No response will be less than 70dB down. The limit at the second harmonic shall be relaxed by 5dB when operated in the high power mode.

3.11 Weight. The RT-1184 shall weigh no more than 23 pounds (see 4.12) less batteries and cold-weather cable.

3.12 Reliability. The RT-1184( )/GRA-114 shall have a specified MTBF (θ^{o}) of 780 hours. The test specified in paragraph 4.14 shall be conducted to determine compliance with this reliability requirement.

3.13 Workmanship. The RT-1184 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.4):

General Requirements for Plastic Material and Plastic Parts
Wiring and Cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, General
- Grounding, general
- Shielding on wire and cable

Soldering and brazing
Cleaning of units
General Requirements for Securing of Parts
Cleaning prior to welding
Controls
Self tapping screws
Cleaning after welding
Printed wiring assemblies

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).
(2) Quality conformance inspection of packaging (see 4.8).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish thirteen (13) first article units of the RT-1184 for Groups A, B, C and D inspections.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in Table VII, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE VII. First article inspection.

	Reqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Units 6 - 13
1. Inspections covered by subsidiary documents 1/	3.3,3.4 3.5,3.15	4.4	Inspection to be performed on all units					
2. Burn-in 1/	3.7	4.7	Inspection to be performed on all units					
3. Group A inspection 1/	See Table VIII		Inspection to be performed on all units					
4. Group B inspection 1/	See Table IX		Inspection to be performed on all units					
5. Group C inspection 2/								
Front-end overload	3.8.2.9	4.9			1			
Duplexing loss	3.8.1.3	4.9			2			
High temperature	3.9.1.1	4.10.1.1		2				
Low temperature	3.9.1.2	4.10.1.2		3				
Humidity	3.9.2	4.10.2		4				
Immersion	3.9.3	4.10.3	2					
Dust	3.9.4	4.10.4			4			
Bounce	3.9.5.1	4.10.5.1			5			
Shock; drop	3.9.5.1	4.10.5.2	3					
Rain	3.9.6	4.10.6			3			
Salt fog	3.9.7	4.10.7				1		
Fungus	3.9.8	4.10.8					1	
EMI	3.10	4.11	1					
Weight	3.11	4.12		1				
6. Group D inspection	3.12	4.14						1

1/ Inspections 1 to 4, shall be performed, in the order shown, on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the unit columns in the table, for inspections in 5, specify the order of inspections for the indicated unit.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4 and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VIII. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table VIII.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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TABLE VIII. Group A inspection.

Inspection	Reqt. Para.	Test Para	AQL Major	AQL Major
Visual and mechanical	3.13	4.13	1%	4%
Electrical				1/
Power output	3.8.1.1	4.9	1%	
Deviation limiting	3.8.1.6	4.9	1%	
Dynamic range	3.8.1.9.1	4.9	1%	
Telemetry operation	3.8.1.10	4.9	1%	
High voltage power supply	3.8.1.11	4.9	1%	
Protective circuit	3.8.1.11.1	4.9	1%	
Ripple	3.8.1.11.2	4.9	1%	
Fail safe timer	3.8.1.12	4.9	1%	
Code sensitivity	3.8.2.1	4.9	1%	
Sensitivity	3.8.2.3	4.9	1%	
Limiting	3.8.2.4	4.9	1%	
Distortion	3.8.2.5	4.9	1%	
Audio output power	3.8.2.7	4.9	1%	
Instrumentation	3.8.3	4.9	1%	

1/ All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IX and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. Group B sample plans, for the AQLs listed in Table IX, shall be as follows:

AQL	Inspection level
4%	S-4
6.5%	S-4

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TABLE IX. Group B inspection.

Inspection	Requirement Paragraph	Test Paragraph	AQL
Interchangeability	3.6	4.6	6.5%
Electrical			
Efficiency	3.8.1.2	4.9	4.0%
Frequency	3.8.1.4	4.9	4.0%
Voice deviation	3.8.1.5	4.9	4.0%
Transmitter voice frequency response	3.8.1.7	4.9	4.0%
Audio distortion	3.8.1.8	4.9	4.0%
Data distortion	3.8.1.9.2	4.9	4.0%
Data frequency response	3.8.1.9.3	4.9	4.0%
Data start up delay	3.8.1.9.4	4.9	4.0%
Code capture ratio	3.8.2.2	4.9	4.0%
Voice frequency response	3.8.2.6	4.9	4.0%
Receiver input power	3.8.2.8	4.9	4.0%
Duplexing desensitization	3.8.2.10	4.0	4.0%
Meter readings	3.8.3.1	4.9	4.0%

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table X. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Five (5) samples of the complete equipment shall be selected at random, one (1) sample for each Subgroup in Table X, from the first one hundred (100) or fraction thereof produced. The EMI and fungus tests shall be performed only on the first production lot. Thereafter, four (4) samples of the complete equipment shall be selected at random from every one hundred (100) units or fraction thereof greater or equal to 50 units, one unit for each subgroup 1 through 4 of Table X. If the fraction is less than 50 the selection of Group C samples for the preceding test shall include units from the remaining fraction.

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TABLE X. Group C inspection.

Inspection	Requirement Paragraph	Test Paragraph
Subgroup 1		
High temperature	3.9.1.1	4.10.1.1
Low temperature	3.9.1.2	4.10.1.2
Humidity	3.9.2	4.10.2
Subgroup 2		
Rain	3.9.6	4.10.6
Dust	3.9.4	4.10.4
Bounce	3.9.5.1	4.10.5.1
Subgroup 3		
EMI	3.10	4.11
Immersion	3.9.3	4.10.3
Shock; drop	3.9.5.1	4.10.5.2
Subgroup 4		
Salt fog 1/	3.9.7	4.10.7
Subgroup 5		
Fungus 1/	3.9.8	4.10.8

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after inspection before proceeding with subsequent testing.

4.5.3.2 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to, and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the test specified in 4.14 and shall be performed on units which have been subjected to and have passed Group A and Group B inspections.

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4.5.5.1 Sampling for group D inspection. Four (4) units shall be selected at random from each successive 6 months production and subjected to the test of para 4.14. Samples selected from each sixth month's production may be selected early in the sixth month in order that testing be completed by the end of the sixth month.

4.5.5.2 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.5.3 Reinspection of Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Burn-in. Each RT-1184 submitted for acceptance shall be subjected to burn-in in accordance with drawing SC-A-678033 (see 3.7).

4.8 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

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4.9 Performance characteristics. The performance characteristics of Receiver-Transmitter RT-1184( )/GRA-114 shall be tested in accordance with the Performance Test Procedure, Drawing SM-A-865230. Paragraph titles in the test procedure correspond to the requirements paragraphs of the specification.

Title	Requirement para.
Power output	3.8.1.1
Efficiency	3.8.1.2
Duplexing loss	3.8.1.3
Frequency	3.8.1.4
Voice deviation	3.8.1.5
Deviation limiting	3.8.1.6
Transmitter voice frequency response	3.8.1.7
Audio distortion	3.8.1.8
Dynamic range	3.8.1.9.1
Data distortion	3.8.1.9.2
Data frequency response	3.8.1.9.3
Data start up delay	3.8.1.9.4
Telemetry operation	3.8.1.10
High voltage power supply	3.8.1.11
Protective circuit	3.8.1.11.1
Ripple	3.8.1.11.2
Fail-safe timer	3.8.1.12
Code sensitivity	3.8.2.1
Code capture ratio	3.8.2.2
Sensitivity	3.8.2.3
Limiting	3.8.2.4
Distortion	3.8.2.5
Voice frequency response	3.8.2.6
Audio output power	3.8.2.7
Receiver input power	3.8.2.8
Front-end overload	3.8.2.9
Duplexing desensitization	3.8.2.10
Meter readings	3.8.3.1

4.10 Service conditions. The RT-1184 shall be tested in accordance with the following conditions:

4.10.1 Temperature.

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4.10.1.1 High temperature. The RT-1184 shall be subjected to the test of Method 501.1 Procedure II of MIL-STD-810. For Step 4, the chamber temperature shall be 71°C(160°F). For Step 7, the chamber temperature shall be adjusted to 65.6°C(150°F). (This includes effects of solar radiation.) (see 3.9.1.1)

4.10.1.2 Low temperature. The RT-1184 shall be subjected to the test of Method 502.1 Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -45.6°C(-50°F), and shall be maintained until temperature stabilization of the test item is reached. Low operating temperature (Step 4) shall be -40°C(-40°F) (see 3.9.1.2).

4.10.2 Humidity. The RT-1184 shall be subjected to the test of Method 507.1 Procedure III of MIL-STD-810. In Step 7 of Procedure III, the chassis shall be removed from its enclosure, or the test item shall be otherwise opened so as to expose to the chamber environment the normally sealed areas. In Step 6 and 8, measurements shall be taken during the last five hours of each test cycle. (see 3.9.2)

4.10.3 Immersion. The equipment shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810. (See 3.9.3)

4.10.4 Dust. The equipment shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810. (See 3.9.4)

4.10.5 Vibration, bounce and shock.

4.10.5.1 Bounce. The RT-1184 shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810. It shall undergo one hour of testing at each of the following temperatures: (1) Room ambient; (2) -45°C, (-50°F); (3) 71°C, (160°F)

At each of these temperatures RT-1184 shall be vibrated for ten minutes on each face.(see 3.9.5.1)

4.10.5.2 Shock; drop. The RT-1184 shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810. (See 3.9.5.1)

4.10.6 Rain. The RT-1184 shall be subjected to the test of Method 506.1, Procedure I of MIL-STD-810. During each 30 minute rain period, the wind shall be maintained for 15 minutes, commencing five minutes after the start of the period as specified in the procedure. The equipment shall be operated during the last 10 minutes of the 30 minute period. The RT-1184 shall be operated with transit cover open and handset connected during its ten minute operation in rain. During other portions of the test cycle, the transit case shall be closed. (See 3.9.6)

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4.10.7 Salt fog. The RT-1184 shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810. (See 3.9.7)

4.10.8 Fungus. The RT-1184 shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810. Gasket sealed or non-hermetic sealed test items shall be opened during exposure and internal areas shall be sprayed with spore suspension as well as external surfaces. Operation of the equipment is not required. (See 3.9.8)

4.11 Electromagnetic compatibility. The RT-1184 shall be tested in accordance with MIL-STD-462 and meet requirements of 3.10.

4.12 Weight. The RT-1184 shall be weighed and shall meet the requirement of 3.11.

4.13 Visual and mechanical inspection. The RT-1184 shall be examined for the applicable defects listed in MIL-STD-252.

4.14 Reliability testing.

4.14.1 First article and quality conformance Group D inspection. To determine compliance with paragraph 3.12 during First Article and Quality Conformance inspection, the equipment shall be tested in accordance with Test Plan XXIV of MIL-STD-781. The specified MTBF for all reliability testing shall be 780 hours.

4.14.2 Failure definition. A failure is defined as the inability of the equipment to perform its required functions as specified in this specification and in the contractor prepared, government approved reliability test plan.

4.14.3 Test conditions. The test shall be conducted under the conditions of Test Level C of MIL-STD-781 with the exception of no vibration. The test shall be conducted in accordance with the reliability test plan required in the contract.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6(2)(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Receiver-Transmitter RT-1184( )/GRA-114 shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Each Receiver-Transmitter RT-1184( )/GRA-114 shall shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged Method IC-1.

5.1.1.4.2 Receiver-Transmitter RT-1184( )/GRA-114. Each RT-1184 shall be individually unit packed method III as follows: place the required components within the space provided and secure the fasteners. Cushion the RT-1184 on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the container. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservation application and unit packaging. Each unit shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.(2)(b)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Receivers-Transmitter RT-1184( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

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5.2.1.1 Level A unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Receiver-Transmitter RT-1184( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 X 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Level B unitization. Palletization shall be required when: quantities per destination comprise a pallet load. A quantity of Receiver-Transmitter RT-1184( )/GRA-114, unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accomodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.3 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356

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6. NOTES

6.1 Intended use. The Receiver-Transmitter RT-1184( )/GRA-114 when used in conjunction with Receiver-Transmitter RT-1183( )/GRA-114 and Receiver-Transmitter RT-1185( )/GRA-114 is intended to provide a voice and data transmission facility as part of a radio data link for sound ranging applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (see section 5).

c. When rough handling test is required.

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular desing for example RT-1184W/GRA-114. The contractor should apply for nomenclature in accordance with applicable clause in the contract (see 1.1).

Custodian:
Army-EL

Preparing activity:
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Project number 1290-A332

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: MIL-R-49112(EL) RECEIVER-TRANSMITTER RT-1184 ( )/GRA-114

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING.

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
	DATE

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